

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_b43bcbe7-b94\agent-a436d439c7825c618.jsonl`  
- SHA-256: `f6f562b0a1f4fb5283928a509a5c15f245fbf14f3c5ad85d33d10a36ee303702`  
- Source modified: `2026-07-02T03:19:23+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_b43bcbe7-b94`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T03:16:59.068Z)

You are an adversarial verifier in a tech-debt audit of the multi-repo workspace at C:/Users/avidu/Projects/khelsutra-guru (repos: badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport; non-git dirs 'Annotation Setup', khelsutra-screencast). Today is 2026-07-02. All repos are at origin head.
Default stance: REFUTE. A round-1 finder claimed (UNVERIFIED):
REPO: rally-annotator
CLAIM: Content script runs a 3 Hz full-DOM shadow-piercing scan on every page the user visits, even with the panel hidden
INVESTIGATION HINTS: Browser-extension content script; check manifest matches + the scan loop, whether it is gated on panel visibility/site allowlist. Severity hinges on whether the extension is meant for all URLs.
VERIFY against the CURRENT working tree: (1) reproduce the defect/debt from actual code at file:line; (2) check git log for a fix that already landed; (3) construct the concrete failure scenario or downgrade severity; (4) propose the correctly-scoped fix with effort S/M/L; (5) flag OWNER_GATED if it needs owner decisions (real cloud spend, product/strategy, counsel, Cloudflare dashboard).
HARD RULES: READ-ONLY ? no file modifications, no installs (pip download/index checks and WebFetch to pypi.org are OK), no test-suite runs. gh read-only OK. sqlite reads OK. CONFIRMED requires you personally observed the defect in current code/data; cite file:line for every claim.

2. user (2026-07-02T03:17:06.948Z)

```
Annotation Setup  
badminton-highlight-indexer  
khelsutra  
khelsutra-screencast  
rally-annotator  
sports-data-collector  
sports-obsreport  
CHANGELOG.md  
CONTRIBUTING.md  
LICENSE  
README.md  
docs  
test  
vlc  
web
```

3. user (2026-07-02T03:17:13.728Z)

(Bash completed with no output)

4. user (2026-07-02T03:17:14.462Z)

3d604bc docs: sync READMEs with current state (multi-agent audit fixes) (#25)
f5ceefc docs(readme): tick live-test ? v1.7.1 verified in real VLC (incl. Indic rendering) (#24)
a0592f6 docs(readme): refresh stale roadmap + tests lines (#23)
ac245ce feat(vlc): complete localization ? status messages + HELP guide (v1.7.1) (#21)
245bf54 feat(vlc): localize dialog chrome into 7 languages (v1.7) (#20)
0a05ecf feat(web): localize panel into kn/hi/es/da/id/te + per-locale screencasts (#19)
5ec1831 feat(web): i18n infrastructure + English catalog (keys-not-prose) (#18)
75541ab docs: localization (i18n) design + locked decisions (#17)
7094d76 docs(readme): use verified Khelsutra site copy + add "About Khelsutra" (#16)
33a4348 docs: frame as Khelsutra sister repo; add design doc + CONTRIBUTING + PR template (#15)

5. user (2026-07-02T03:17:20.561Z)

./test/dialog_test.lua
./vlc/rally_annotator.lua
./web/package-lock.json
./web/package.json
./web/src/i18n/locales/da/common.json
./web/src/i18n/locales/en/common.json
./web/src/i18n/locales/es/common.json
./web/src/i18n/locales/hi/common.json
./web/src/i18n/locales/id/common.json
./web/src/i18n/locales/kn/common.json
./web/src/i18n/locales/te/common.json
./web/tsconfig.json

6. user (2026-07-02T03:17:29.014Z)

./web/.gitignore
./web/DESIGN.md
./web/e2e/extension.spec.ts
./web/e2e/fixtures/index.html
./web/e2e/fixtures/sample.webm
./web/e2e/serve.mjs
./web/entrypoints/background.ts
./web/entrypoints/content.ts
./web/package-lock.json
./web/package.json
./web/playwright.config.ts
./web/README.md
./web/src/i18n/index.ts
./web/src/i18n/locales/da/common.json
./web/src/i18n/locales/en/common.json
./web/src/i18n/locales/es/common.json
./web/src/i18n/locales/hi/common.json
./web/src/i18n/locales/id/common.json
./web/src/i18n/locales/kn/common.json
./web/src/i18n/locales/te/common.json
./web/src/messages.ts
./web/src/persist/store.ts
./web/src/state/annotator.ts
./web/src/state/csv.ts
./web/src/ui/panel.ts
./web/src/version.ts
./web/src/video/directVideo.ts
./web/test/annotator.test.ts
./web/test/csv.test.ts
./web/test/directVideo.test.ts
./web/test/i18n.test.ts
./web/test/panel.smoke.test.ts
./web/test/store.test.ts
./web/TESTING.md
./web/tsconfig.json
./web/vitest.config.ts
./web/wxt.config.ts

```

---
{
  "name": "rally-annotator-web",
  "version": "0.1.0",
  "private": true,
  "type": "module",
  "description": "Browser-extension port of the Rally Annotator VLC tool (Chrome-first, Manifest
V3, cross-browser-ready).",
  "scripts": {
    "dev": "wxt",
    "build": "wxt build",
    "build:firefox": "wxt build -b firefox",
    "zip": "wxt zip",
    "prepare:wxt": "wxt prepare",
    "test": "vitest run",
    "test:watch": "vitest",
    "coverage": "vitest run --coverage",
    "e2e": "playwright test",
    "e2e:install": "playwright install chromium",
    "typecheck": "tsc --noEmit"
  },
  "devDependencies": {
    "@playwright/test": "^1.61.0",
    "@vitest/coverage-v8": "^3.2.6",
    "jsdom": "^29.1.1",
    "typescript": "^5.7.3",
    "vitest": "^3.2.4",
    "wxt": "^0.20.26"
  }
}

```

7. user (2026-07-02T03:17:34.381Z)

```

1      import { defineConfig } from "wxt";
2
3      // One MV3 source tree; WXT emits per-browser bundles (Chrome now; Firefox/Safari are
4      // the north-star targets ? `wxt build -b firefox` etc.) and manages the background
5      // service-worker vs event-page split per browser. Version comes from package.json.
6      export default defineConfig({
7        manifest: {
8          name: "Rally Annotator",
9          description:
10            "Mark rally start/end + a point-ending reason to a CSV while watching web video "
11            +
12            "(HTML5 <video>, incl. youtube.com/watch). Net-separated racquet sports.",
13          permissions: ["storage", "downloads"],
14          host_permissions: ["<all_urls>"],
15          action: { default_title: "Toggle Rally Annotator panel" },
16        },
17      });

```

8. user (2026-07-02T03:17:34.429Z)

```

1      import { browser } from "wxt/browser";
2      import { Annotator } from "../src/state/annotator";
3      import { DirectVideoHandler } from "../src/video/directVideo";
4      import { deriveIdentity, loadRows, saveRows, csvFilename } from "../src/persist/store";
5      import { mountPanel } from "../src/ui/panel";
6      import type { RuntimeMessage } from "../src/messages";
7
8      // Top-frame content script: derives the per-video identity, loads any saved rallies,
9      // wires the pure Annotator to the page's video + storage + CSV download, and mounts the
10     // panel (hidden until the toolbar icon is clicked). Cross-origin EMBEDDED players
11     // (youtube.com/embed inside a third-party page) are a later, separate handler.

```

```

12 export default defineContentScript({
13   matches: ["<all_urls>"],
14   runAt: "document_idle",
15   async main() {
16     if (window.top !== window) return; // v1: top frame only
17
18     const identity = deriveIdentity();
19     const video = new DirectVideoHandler();
20     const rows = await loadRows(identity.key);
21
22     const annotator = new Annotator({
23       rows,
24       // chrome.storage.local is the source of truth (extension-scoped). Async mirror;
25       // the in-memory annotator.rows is authoritative for the live session.
26       persist: (rs) => {
27         void saveRows(identity.key, rs);
28         return { ok: true };
29       },
30     });
31
32     const panel = mountPanel({
33       annotator,
34       video,
35       identity,
36       download: (csv) =>
37         void browser.runtime.sendMessage({
38           type: "rally-download",
39           filename: csvFilename(identity.title),
40           csv,
41         } as RuntimeMessage),
42     });
43
44     panel.toggle(); // start hidden; the toolbar icon reveals it on pages you want to
45     annotate
46
47     browser.runtime.onMessage.addListener((msg: RuntimeMessage) => {
48       if (msg && msg.type === "toggle-panel") panel.toggle();
49     });
50   },
51 });

```

9. user (2026-07-02T03:17:39.836Z)

```

1 // Finds and controls the active HTML5 <video> on the page ? including videos nested
2 // in (open or closed) shadow DOM ? and exposes the playback primitives the annotator
3 // needs: the web analogue of VLC's now_seconds() / seek_by() / play_pause().
4 //
5 // Key simplification vs VLC: HTML5 video.currentTime is already in SECONDS, so there
6 // is no microsecond conversion. This handler covers same-origin/top-frame videos
7 // (including youtube.com/watch, whose player is in the main page). Cross-origin
8 // EMBEDDED players (e.g. a youtube.com/embed iframe on a third-party site) are a
9 // separate, later handler ? they live in a frame this top-frame script can't reach.
10
11 // chrome.dom.openOrClosedShadowRoot lets a content script pierce CLOSED shadow roots
12 // (Chrome/Firefox); fall back to the public .shadowRoot otherwise.
13 function openOrClosed(el: Element): ShadowRoot | null {
14   const anyChrome = (globalThis as unknown as { chrome?: any }).chrome;
15   try {
16     if (anyChrome?.dom?.openOrClosedShadowRoot) {
17       return anyChrome.dom.openOrClosedShadowRoot(el) ?? null;
18     }
19   } catch {
20     /* ignore */
21   }

```

```

22     return (el as unknown as { shadowRoot?: ShadowRoot | null }).shadowRoot ?? null;
23 }
24
25 function collectVideos(root: ParentNode, acc: HTMLVideoElement[], depth: number): void {
26     if (depth > 8) return; // bound the shadow-DOM walk
27     root.querySelectorAll("video").forEach((v) => acc.push(v as HTMLVideoElement));
28     root.querySelectorAll("*").forEach((el) => {
29         const sr = openOrClosed(el as Element);
30         if (sr) collectVideos(sr, acc, depth + 1);
31     });
32 }
33
34 export class DirectVideoHandler {
35     private current: HTMLVideoElement | null = null;
36
37     findActive(): HTMLVideoElement | null {
38         const vids: HTMLVideoElement[] = [];
39         collectVideos(document, vids, 0);
40         if (vids.length === 0) {
41             this.current = null;
42             return null;
43         }
44         // Prefer a currently-playing video; tie-break by largest rendered area.
45         const score = (v: HTMLVideoElement) => {
46             const r = v.getBoundingClientRect();
47             const area = Math.max(0, r.width) * Math.max(0, r.height);
48             return (v.paused ? 0 : 1e12) + area;
49         };
50         vids.sort((a, b) => score(b) - score(a));
51         this.current = vids[0];
52         return this.current;
53     }
54
55     private active(): HTMLVideoElement | null {
56         if (this.current && this.current.isConnected) return this.current;
57         return this.findActive();
58     }
59
60     hasVideo(): boolean {
61         return this.active() != null;
62     }
63
64     now(): number | null {
65         const v = this.active();
66         return v ? v.currentTime : null;
67     }
68
69     duration(): number | null {
70         const v = this.active();
71         return v && Number.isFinite(v.duration) ? v.duration : null;
72     }
73
74     isPaused(): boolean | null {
75         const v = this.active();
76         return v ? v.paused : null;
77     }
78
79     seekBy(delta: number): boolean {
80         const v = this.active();
81         if (!v) return false;
82         let t = v.currentTime + delta;
83         if (t < 0) t = 0;
84         v.currentTime = t;
85         return true;
86     }

```

```

87
88 // Single toggle gated on actual paused state (mirrors the Lua play_pause()).
89 playPause(): "playing" | "paused" | null {
90     const v = this.active();
91     if (!v) return null;
92     if (v.paused) {
93         void v.play().catch(() => {});
94         return "playing";
95     }
96     v.pause();
97     return "paused";
98 }
99 }
100

```

10. user (2026-07-02T03:17:40.254Z)

```

1 // In-page annotation panel ? the browser analogue of the VLC dialog. Rendered into an
2 // OPEN Shadow DOM so the host page's CSS can't bleed in (and vice-versa) while still
3 // being reachable by automation/devtools ("closed" is not a real security boundary).
4 // Every user-facing string goes through the i18n shim t() (keys-not-prose);
reason/sport
5 // VALUES stay canonical (written to the CSV) ? only their display labels are
translated.
6
7 import { Annotator, SPORTS, REASON_OPTIONS, type Msg } from "../state/annotator";
8 import type { DirectVideoHandler } from "../video/directVideo";
9 import type { VideoIdentity } from "../persist/store";
10 import { VERSION } from "../version";
11 import { t, getLocale, setLocale, SUPPORTED_LOCALES, LOCALE_LABELS, type Locale } from
"../i18n";
12
13 export interface PanelDeps {
14     annotator: Annotator;
15     video: DirectVideoHandler;
16     identity: VideoIdentity;
17     download: (csv: string) => void;
18     onLocaleChange?: (locale: Locale) => void; // optional: persist the choice (content
script wires this)
19 }
20
21 export interface PanelHandle {
22     toggle(): void;
23     destroy(): void;
24 }
25
26 // mm:ss.mmm for display (matches the Lua fmt_clock).
27 function fmtClock(s: number | null): string {
28     if (s == null) return "--:--";
29     if (s < 0) s = 0;
30     const m = Math.floor(s / 60);
31     const sec = s - m * 60;
32     return `${m}:${sec.toFixed(3).padStart(6, "0")}`;
33 }
34
35 function esc(t: string): string {
36     return t.replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;");
37 }
38
39 // Translate a Msg, mapping canonical reason/sport params to their display labels.
40 function tMsg(m: Msg): string {
41     let p = m.params;
42     if (p && (typeof p.reason === "string" || typeof p.sport === "string")) {
43         p = { ...p };
44         if (typeof p.reason === "string") p.reason = t("reason." + p.reason);
45     }
46 }
47

```

```

45     if (typeof p.sport === "string") p.sport = t("sport." + p.sport);
46   }
47   return t(m.key, p);
48 }
49
50 const STYLE = `
51 :host { all: initial; }
52 * { box-sizing: border-box; font-family: system-ui, "Noto Sans", "Noto Sans Devanagari",
53     "Noto Sans Kannada", "Noto Sans Telugu", "Nirmala UI", sans-serif; }
54 .card {
55   position: fixed; top: 16px; right: 16px; width: 340px; z-index: 2147483647;
56   background: #1e1f22; color: #e6e6e6; border: 1px solid #444; border-radius: 8px;
57   box-shadow: 0 8px 28px rgba(0,0,0,.5); font-size: 12px; user-select: none;
58 }
59 .hdr { display: flex; align-items: center; gap: 6px; padding: 7px 9px; cursor: move;
60   background: #2b2d31; border-radius: 8px 8px 0 0; }
61 .hdr .t { font-weight: 600; flex: 1; }
62 .hdr button { cursor: pointer; }
63 .body { padding: 9px; display: flex; flex-direction: column; gap: 7px; }
64 .row { display: flex; gap: 6px; align-items: center; }
65 .row > * { min-width: 0; }
66 label { color: #aab; white-space: nowrap; }
67 input, select, button {
68   font-size: 12px; background: #2b2d31; color: #e6e6e6; border: 1px solid #555;
69   border-radius: 5px; padding: 4px 6px;
70 }
71 input { width: 100%; }
72 button { cursor: pointer; background: #3a3d44; }
73 button:hover { background: #474b54; }
74 button.primary { background: #2f6f4f; border-color: #3c8a63; }
75 button.primary:hover { background: #38805c; }
76 .grid2 { display: grid; grid-template-columns: 1fr 1fr; gap: 6px; }
77 .status { background: #15161a; border: 1px solid #333; border-radius: 5px; padding: 6px;
78   line-height: 1.45; min-height: 52px; color: #cfd3da; }
79 .list { background: #15161a; border: 1px solid #333; border-radius: 5px; height: 122px;
80   overflow-y: auto; }
81 .list .item { padding: 3px 6px; cursor: pointer; border-bottom: 1px solid #232427;
82   white-space: nowrap; }
83 .list .item:hover { background: #25262b; }
84 .list .item.sel { background: #2f4a6f; }
85 .help { background: #15161a; border: 1px solid #333; border-radius: 5px; padding: 7px;
86   line-height: 1.5; max-height: 260px; overflow-y: auto; color: #cfd3da; }
87 .mini { font-size: 11px; color: #8b90a0; }
88 `;
89
90 function h<K extends keyof HTMLElementTagNameMap>(  

91   tag: K,  

92   props: Partial<HTMLElementTagNameMap[K]> & { class?: string } = {},  

93   ...kids: (Node | string)[]  

94 ): HTMLElementTagNameMap[K] {  

95   const e = document.createElement(tag);  

96   for (const [k, v] of Object.entries(props)) {  

97     if (k === "class") e.className = v as string;  

98     else (e as any)[k] = v;  

99   }  

100   for (const kid of kids) e.append(kid as any);  

101   return e;  

102 }
103
104 export function mountPanel(deps: PanelDeps): PanelHandle {  

105   const { annotator: a, video, download } = deps;  

106  

107   const host = document.createElement("div");  

108   host.style.all = "initial";  

109   const root = host.attachShadow({ mode: "open" });

```

```

110     root.append(h("style", { textContent: STYLE }));
111
112     const card = h("div", { class: "card" });
113     root.append(card);
114
115     // ---- header (drag handle) ----
116     const titleEl = h("span", { class: "t" });
117     const helpBtn = h("button", { title: "Toggle help" });
118     const hideBtn = h("button", { textContent: "?", title: "Hide panel" });
119     const hdr = h("div", { class: "hdr" }, titleEl, helpBtn, hideBtn);
120     card.append(hdr);
121
122     const body = h("div", { class: "body" });
123     card.append(body);
124
125     // ---- language ----
126     const langLabel = h("label");
127     const langSel = h("select", { name: "language" }) as HTMLSelectElement;
128     for (const l of SUPPORTED_LOCALES) langSel.append(h("option", { value: l, textContent:
LOCALE_LABELS[l] }));
129     langSel.value = getLocale();
130     body.append(h("div", { class: "row" }, langLabel, langSel));
131
132     // ---- sport ----
133     const sportLabel = h("label");
134     const sportSel = h("select", { name: "sport" }) as HTMLSelectElement;
135     for (const s of SPORTS) sportSel.append(h("option", { value: s }));
136     sportSel.value = a.sport;
137     body.append(h("div", { class: "row" }, sportLabel, sportSel));
138
139     // ---- playback ----
140     const backBtn = h("button");
141     const playBtn = h("button");
142     const fwdBtn = h("button");
143     body.append(h("div", { class: "row" }, backBtn, playBtn, fwdBtn));
144
145     // ---- start / end ----
146     const startLabel = h("label");
147     const startIn = h("input", { name: "start" }) as HTMLInputElement;
148     const endLabel = h("label");
149     const endIn = h("input", { name: "end" }) as HTMLInputElement;
150     body.append(h("div", { class: "row" }, startLabel, startIn, endLabel, endIn));
151
152     // ---- next # / shots ----
153     const nextLabel = h("label");
154     const nextIn = h("input", { name: "next" }) as HTMLInputElement;
155     const shotsLabel = h("label");
156     const shotsIn = h("input", { name: "shots" }) as HTMLInputElement;
157     body.append(h("div", { class: "row" }, nextLabel, nextIn, shotsLabel, shotsIn));
158
159     // ---- reason ----
160     const reasonLabel = h("label");
161     const reasonSel = h("select", { name: "reason" }) as HTMLSelectElement;
162     for (const r of REASON_OPTIONS) reasonSel.append(h("option", { value: r }));
163     body.append(h("div", { class: "row" }, reasonLabel, reasonSel));
164
165     // ---- mark / save ----
166     const markStartBtn = h("button");
167     const markEndBtn = h("button");
168     const saveBtn = h("button", { class: "primary" });
169     body.append(h("div", { class: "grid2" }, markStartBtn, markEndBtn));
170     body.append(h("div", { class: "row" }, saveBtn));
171
172     // ---- status ----
173     const status = h("div", { class: "status" });

```



```

174     body.append(status);
175
176     // ---- help (hidden by default) ----
177     const help = h("div", { class: "help" });
178     help.style.display = "none";
179     body.append(help);
180
181     // ---- recent list ----
182     const recentLabel = h("div", { class: "mini" });
183     body.append(recentLabel);
184     const list = h("div", { class: "list" });
185     body.append(list);
186
187     // ---- actions ----
188     const editBtn = h("button");
189     const delBtn = h("button");
190     const undoBtn = h("button");
191     const refreshBtn = h("button");
192     const dlBtn = h("button");
193     body.append(h("div", { class: "grid2" }, editBtn, delBtn));
194     body.append(h("div", { class: "grid2" }, undoBtn, refreshBtn));
195     body.append(h("div", { class: "row" }, dlBtn));
196
197     // ---- state ----
198     let selectedN: number | null = null;
199     let lastMsg: Msg = { key: "status.ready" };
200
201     function setStatus(m: Msg) {
202         lastMsg = m;
203         render();
204     }
205
206     // Language-only text (static labels, option labels, placeholders, fixed buttons,
help).
207     // Re-applied on mount and whenever the locale changes.
208     function applyI18n() {
209         titleEl.textContent = `? ${t("panel.title")} v${VERSION}`;
210         langLabel.textContent = t("label.language");
211         sportLabel.textContent = t("label.sport");
212         for (const o of Array.from(sportSel.options)) o.textContent = t("sport." + o.value);
213         backBtn.textContent = t("btn.back5");
214         playBtn.textContent = t("btn.playPause");
215         fwdBtn.textContent = t("btn.fwd5");
216         startLabel.textContent = t("label.start");
217         endLabel.textContent = t("label.end");
218         nextLabel.textContent = t("label.next");
219         shotsLabel.textContent = t("label.shots");
220         startIn.placeholder = t("ph.start");
221         endIn.placeholder = t("ph.end");
222         nextIn.placeholder = t("ph.next");
223         shotsIn.placeholder = t("ph.shots");
224         reasonLabel.textContent = t("label.reason");
225         for (const o of Array.from(reasonSel.options)) o.textContent = t("reason." +
o.value);
226         editBtn.textContent = t("btn.edit");
227         delBtn.textContent = t("btn.delete");
228         refreshBtn.textContent = t("btn.refresh");
229         dlBtn.textContent = t("btn.downloadCsv");
230         recentLabel.textContent = t("label.recent");
231         help.innerHTML = t("help.html");
232     }
233
234     function render() {
235         // fields
236         startIn.value = a.startField;

```

```

237     endIn.value = a.endField;
238     nextIn.value = a.nextField;
239     shotsIn.value = a.shotsField;
240     reasonSel.value = a.reason;
241     sportSel.value = a.sport;
242     langSel.value = getLocale();
243     // state-dependent labels
244     markStartBtn.textContent = tMsg(a.markStartLabel());
245     markEndBtn.textContent = tMsg(a.markEndLabel());
246     saveBtn.textContent = tMsg(a.saveLabel());
247     undoBtn.textContent = tMsg(a.undoLabel());
248     helpBtn.textContent = help.style.display === "none" ? t("btn.help") :
t("btn.hideHelp");
249     // status
250     const editing = a.mode === "edit" && a.editIndex !== null ? a.rows[a.editIndex] :
null;
251     const last = a.lastRow();
252     const lines = [
253       tMsg(lastMsg),
254       editing ? t("status.modeEdit", { n: editing.n }) : t("status.modeNew"),
255     ];
256     if (last) {
257       lines.push(
258         tMsg({
259           key: "status.lastRow",
260           params: { n: last.n, s: fmtClock(last.s), e: fmtClock(last.e), reason:
last.reason },
261         })
262       );
263     }
264     lines.push(t("status.footer", { now: fmtClock(video.now()), count: a.rows.length
}));
265     status.innerHTML =
266       lines.map(esc).join("<br>") +
267       `<br><span class="mini">${esc(t("status.csvLine", { key: deps.identity.key
})))}</span>`;
268     // list
269     list.replaceChildren();
270     for (const r of a.rows) {
271       const shotsTxt = r.shots !== null ? ` ${r.shots} ${t("ph.shots")}` : "";
272       const item = h("div", {
273         class: "item" + (r.n === selectedN ? " sel" : ""),
274        .textContent: `#${r.n} ${fmtClock(r.s)} ? ${fmtClock(r.e)} [${t("reason." +
r.reason)}, ${t("sport." + r.sport)}]${shotsTxt}`,
275       });
276       item.addEventListener("click", () => {
277         selectedN = r.n;
278         render();
279       });
280       list.append(item);
281     }
282   }
283
284   // ---- wiring ----
285   startIn.addEventListener("input", () => {
286     a.startField = startIn.value;
287     render();
288   });
289   endIn.addEventListener("input", () => {
290     a.endField = endIn.value;
291     render();
292   });
293   nextIn.addEventListener("input", () => {
294     a.nextField = nextIn.value;
295     render();

```

```

296     });
297     shotsIn.addListener("input", () => {
298         a.shotsField = shotsIn.value;
299     });
300     reasonSel.addListener("change", () => {
301         a.reason = reasonSel.value;
302     });
303     sportSel.addListener("change", () => {
304         a.sport = sportSel.value;
305     });
306     langSel.addListener("change", () => {
307         setLocale(langSel.value as Locale);
308         deps.onLocaleChange?.(getLocale());
309         applyI18n();
310         render();
311     });
312
313     backBtn.addListener("click", () =>
314         setStatus({ key: video.seekBy(-5) ? "status.seekBack" : "status.noVideoSeek" })
315     );
316     fwdBtn.addListener("click", () =>
317         setStatus({ key: video.seekBy(5) ? "status.seekFwd" : "status.noVideoSeek" })
318     );
319     playBtn.addListener("click", () => {
320         const st = video.playPause();
321         setStatus({ key: st === "playing" ? "status.resumed" : st === "paused" ?
"status.paused" : "status.noVideoLoaded" });
322     });
323
324     markStartBtn.addListener("click", () => setStatus(a.markStart(video.now())));
325     markEndBtn.addListener("click", () => setStatus(a.markEnd(video.now())));
326     saveBtn.addListener("click", () => {
327         const res = a.saveRally();
328         if (res.ok) download(a.toCSV()); // download the full CSV on each save (per design)
329         setStatus(res);
330     });
331
332     editBtn.addListener("click", () => setStatus(a.editSelected(selectedN)));
333     delBtn.addListener("click", () => {
334         const res = a.deleteSelected(selectedN);
335         if (res.ok) selectedN = null;
336         setStatus(res);
337     });
338     undoBtn.addListener("click", () => setStatus(a.undoLast()));
339     refreshBtn.addListener("click", () => {
340         video.findActive();
341         setStatus({ key: video.hasVideo() ? "status.refreshedVideo" : "status.noVideoFound"
});
342     });
343     dlBtn.addListener("click", () => {
344         download(a.toCSV());
345         setStatus({ key: "status.downloaded", params: { count: a.rows.length } });
346     });
347
348     helpBtn.addListener("click", () => {
349         help.style.display = help.style.display === "none" ? "block" : "none";
350         render();
351     });
352     hideBtn.addListener("click", () => api.toggle());
353
354     // ---- dragging ----
355     let drag: { x: number; y: number; left: number; top: number } | null = null;
356     hdr.addListener("mousedown", (e) => {
357         const rect = card.getBoundingClientRect();
358         card.style.right = "auto";

```

```

359     card.style.left = rect.left + "px";
360     card.style.top = rect.top + "px";
361     drag = { x: e.clientX, y: e.clientY, left: rect.left, top: rect.top };
362     e.preventDefault();
363   });
364   const onMove = (e: MouseEvent) => {
365     if (!drag) return;
366     card.style.left = drag.left + (e.clientX - drag.x) + "px";
367     card.style.top = drag.top + (e.clientY - drag.y) + "px";
368   };
369   const onUp = () => {
370     drag = null;
371   };
372   window.addEventListener("mousemove", onMove);
373   window.addEventListener("mouseup", onUp);
374
375   // ---- live clock ----
376   const clock = window.setInterval(render, 333);
377
378   // ---- fullscreen re-parent (so the panel survives fullscreen video; desktop) ----
379   const onFs = () => {
380     const fe = document.fullscreenElement;
381     if (fe && !fe.contains(host)) fe.append(host);
382     else if (!fe && host.parentElement !== document.body) document.body.append(host);
383   };
384   document.addEventListener("fullscreenchange", onFs);
385
386   document.body.append(host);
387   applyI18n();
388   render();
389
390   const api: PanelHandle = {
391     toggle() {
392       host.style.display = host.style.display === "none" ? "" : "none";
393     },
394     destroy() {
395       window.clearInterval(clock);
396       window.removeEventListener("mousemove", onMove);
397       window.removeEventListener("mouseup", onUp);
398       document.removeEventListener("fullscreenchange", onFs);
399       host.remove();
400     },
401   };
402   return api;
403 }
404

```

11. user (2026-07-02T03:18:07.802Z)

0a05ecf feat(web): localize panel into kn/hi/es/da/id/te + per-locale screencasts (#19)
5ec1831 feat(web): i18n infrastructure + English catalog (keys-not-prose) (#18)
ad6ac9c Add rally-annotator web extension (Chrome MV3) with proof-driven tests + CI (#14)

- * main
 - remotes/origin/claude/intelligent-planck-fokdq2
 - remotes/origin/feat/layout-snapshot-ci
 - remotes/origin/feat/version-in-window-title
 - remotes/origin/fix/gitattributes-eol-lf
 - remotes/origin/main

12. user (2026-07-02T03:18:08.942Z)

C:\Users\avidu\Projects\khelesutra-guru\rally-annotator\web\src\ui\panel.ts:376: const clock = window.setInterval(render, 333);

13. user (2026-07-02T03:18:16.007Z)

```
1      import { browser } from "wxt/browser";
2      import type { RuntimeMessage } from "../src/messages";
3
4      // Background service worker: performs the CSV download (chrome.downloads is unavailable
5      // to content scripts) and routes the toolbar-icon click into a panel toggle.
6      export default defineBackground(() => {
7        browser.runtime.onMessage.addListener((msg: RuntimeMessage) => {
8          if (msg && msg.type === "rally-download") {
9            const url = "data:text/csv;charset=utf-8," + encodeURIComponent(msg.csv);
10            void browser.downloads
11              .download({ url, filename: msg.filename, saveAs: false, conflictAction:
"overwrite" })
12              .catch((e) => console.error("[rally] download failed:", e));
13          }
14        });
15
16        // No default_popup is set, so the icon click fires onClicked -> toggle the panel.
17        browser.action.onClicked.addListener((tab) => {
18          if (tab.id !== null) {
19            void browser.tabs
20              .sendMessage(tab.id, { type: "toggle-panel" } as RuntimeMessage)
21              .catch(() => {
22                /* tab has no content script (e.g. chrome:// page) */
23              });
24          }
25        });
26      });
27
```

14. user (2026-07-02T03:18:17.603Z)

112: **Why.** Auto-showing a panel on every page (the content script matches `<all_urls>`) is intrusive. The

15. user (2026-07-02T03:18:23.591Z)

Consequence. ~10 MB default cap (fine here). If storage ever needs to scale or be queryable, **IndexedDB** in the service worker (message-passed) is the documented escalation path.

LD-5 ? UI: open Shadow DOM

Decision. Render the panel in an **open** shadow root.
Why. Open and closed give **identical CSS isolation**; "closed" is not a real security boundary (the host page can hook `attachShadow`). Open keeps the panel reachable by automation/devtools ? which is what makes the E2E (LD-8) possible.

LD-6 ? Time base: seconds (no microseconds)

Decision. Read/write `video.currentTime` directly (float seconds); CSV times stay 3-dp seconds.
Why. HTML5 media time is already seconds, unlike VLC's microseconds ? the conversion simply disappears.
Note (verified). Some UAs reduce `currentTime` precision (Firefox ~2 ms by default); don't assume sub-ms.

LD-7 ? Panel hidden by default; toolbar icon toggles

Decision. The content script mounts the panel **hidden**; clicking the extension's toolbar icon toggles it (routed via the background SW).
Why. Auto-showing a panel on every page (the content script matches `<all_urls>`) is intrusive. The icon is an explicit "annotate this page" gesture.

LD-8 ? Testing is proof-and-validation driven; it is the merge gate

Decision. Unit + jsdom tests with **enforced coverage thresholds**, **plus** a real-Chromium Playwright E2E that loads the built extension and records a screencast, **plus** CI that runs all of it.

A PR merges only when this is green.

Why. "It works" must be evidence, not a claim ? especially so outside contributors' PRs can be

trusted. Browser-only behavior is proven by the E2E rather than asserted. See

TESTING.md.

LD-9 ? Safari is App-Store-only (deferred)

Decision (verified). Safari Web Extensions reach end users **only via the App Store** (macOS + iOS) ?

there is no Developer ID direct-install. They require a Mac + `xcrun safari-web-extension-converter`.

Consequence. Safari cannot be built or E2E-tested on Windows or default CI runners; it is documented, not implemented.

LD-12 ? Localization (i18n) mirrors Khelsutra

Decision. The panel UI is localized into Khelsutra's locales **+ Telugu** (`en, kn, hi, es, da, id,

te`) via a dependency-free `t()` shim over a shared `common.json` catalog, mirroring Khelsutra's i18n

foundation (keys-not-prose, ICU-lite, always-fall-back-to-`en`, key-parity + no-bare-strings CI gates,

OFL Noto fonts, AI-draft ? native-review). The reason/sport **values** and CSV header stay canonical

English (LD-3 byte-compatibility is preserved across all languages). Full design + locked decisions:

../docs/LOCALIZATION.md.

16. user (2026-07-02T03:18:39.541Z)

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\test\panel.smoke.test.ts:50:
```

```
panel!.destroy();
```

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\test\panel.smoke.test.ts:68:
```

```
panel!.destroy();
```

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\test\panel.smoke.test.ts:79:
```

```
panel!.destroy();
```

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\test\panel.smoke.test.ts:116:
```

```
panel!.destroy();
```

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\test\panel.smoke.test.ts:147:
```

```
panel!.destroy();
```

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\test\panel.smoke.test.ts:153:
```

```
panel!.destroy();
```

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\src\ui\panel.ts:23:  destroy(): void;
```

```
C:\Users\avidu\Projects\khelsutra-guru\rally-annotator\web\src\ui\panel.ts:394:    destroy() {
```

17. assistant (2026-07-02T03:19:05.328Z)

All elements of the claim verified in current code. Compiling the verdict.

18. user (2026-07-02T03:19:23.704Z)

Structured output provided successfully